

TidyTuesday Week 37

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Tuesday 8th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the corner of the internet where we throw two economic time series into a blender, hit "regress," and pretend to be surprised at whatever chunky smoothie comes out. This week our little machine served up only two matchups, but boy did it commit to the bit, pairing things that have absolutely no business being in the same spreadsheet, let alone the same regression.

First up, Monday's contestant: the "Discussion About Pandemics Index for Lithuania" versus "Means of Financing: Borrowing from the Public." Yes, we regressed how much Lithuanians are chatting about pandemics against how much the U.S. government is borrowing. The raw regression came back significant (p-value around 0.0002, R-squared of about 0.20), and I want you to sit with the profound global insight there for exactly zero seconds. The genuinely amusing part is that this one actually survived detrending – p-value still tiny (~ 0.0005), R-squared barely budging to 0.17. So even after we strip out each series' natural drift over time, the relationship stubbornly hangs around. Is there a deep truth here about pandemic anxiety in the Baltics moving in lockstep with American public debt? No. Absolutely not. But it's the rarer kind of result that doesn't instantly evaporate, so I'm legally obligated to raise an eyebrow before returning to my regularly scheduled skepticism.

Then Tuesday gave us the real heavyweight: U.S. carbon dioxide emissions from coal versus the "Congenital Anomalies Price Index" from medical expenditure data. The raw regression positively strutted onto the stage with an R-squared of 0.61 – a number so high that in this project it should immediately be treated as a suspect, not a celebrity. And yet, plot twist, this one also holds up under detrending, actually posting a bigger coefficient once the trends are removed (R-squared still a hefty 0.56, p-value ~ 0.0001). Two series with different start dates, different everything, and they keep dancing together even after we take away the music. My working theory is that the universe is simply messing with me, which is honestly the most rigorous hypothesis on offer.

As always, the mandatory reality check: these are randomly paired series, thrown together by a script that has no idea what any of these words mean, and neither the raw nor the detrended results imply that anything causes anything. Coal emissions are not setting medical price indexes, and Lithuanian pandemic chatter is not printing Treasury debt. Spurious correlation isn't the exception around here – it's basically the house band. This week just happened to produce two results that refused to fully collapse when poked, which is less "scientific breakthrough" and more "statistical funhouse mirror." Enjoy responsibly, cite nothing, and I'll see you next week when the

machine inevitably correlates cheese prices with sunspots.

2 Date: 2026-09-07

Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MTSMFBP133FMS	R-squared:	0.199			
Model:	OLS	Adj. R-squared:	0.186			
Method:	Least Squares	F-statistic:	15.87			
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000176			
Time:	16:43:46	Log-Likelihood:	-893.46			
No. Observations:	66	AIC:	1791.			
Df Residuals:	64	BIC:	1795.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.886e+04	2.46e+04	1.985	0.051	-321.644	9.8e+04
value_fred_WPDILTU	1205.6312	302.613	3.984	0.000	601.092	1810.170
Omnibus:	51.230	Durbin-Watson:	2.093			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	275.412			
Skew:	2.139	Prob(JB):	1.57e-60			
Kurtosis:	12.047	Cond. No.	87.5			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

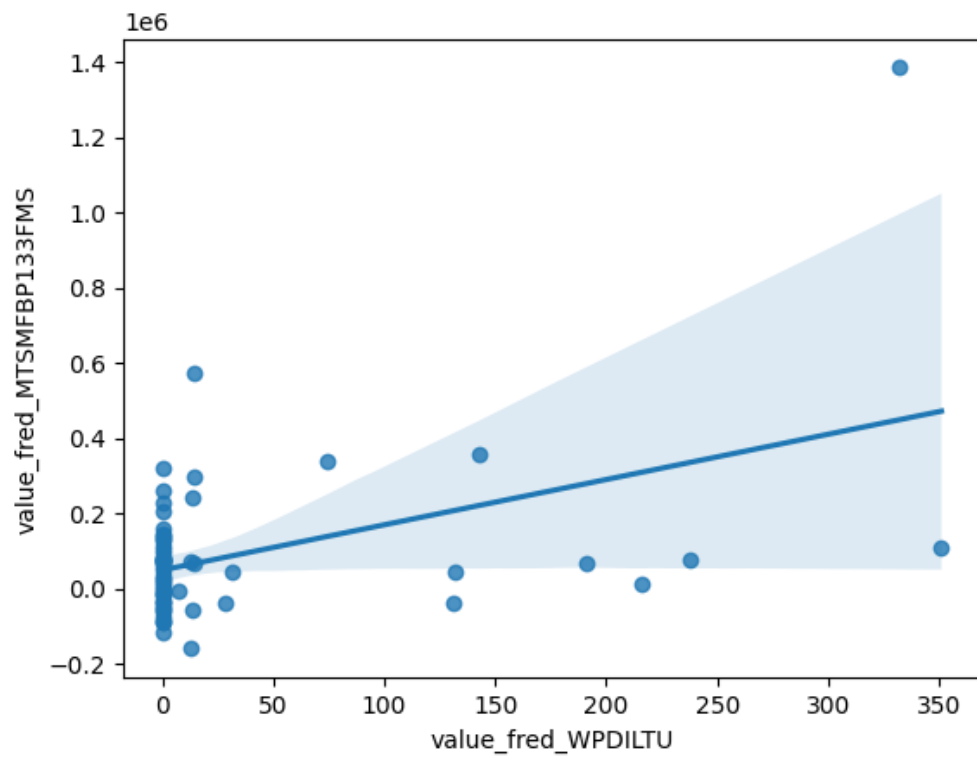


Figure 1: Raw Regression Plot for 2026-09-07

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MTSMFBP133FMS_detrended	R-squared:	0.174
Model:	OLS	Adj. R-squared:	0.161
Method:	Least Squares	F-statistic:	13.48
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000493
Time:	16:43:46	Log-Likelihood:	-893.06
No. Observations:	66	AIC:	1790.
Df Residuals:	64	BIC:	1794.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	5.051e-09	2.28e+04	2.22e-13	1.000	-4.55e+04	4.55e+04
value_fred_WPDILTU_detrended	1138.7087	310.105	3.672	0.000	519.204	1758.214

Omnibus:	52.012	Durbin-Watson:	2.129
Prob(Omnibus):	0.000	Jarque-Bera (JB):	302.597
Skew:	2.138	Prob(JB):	1.96e-66
Kurtosis:	12.579	Cond. No.	73.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

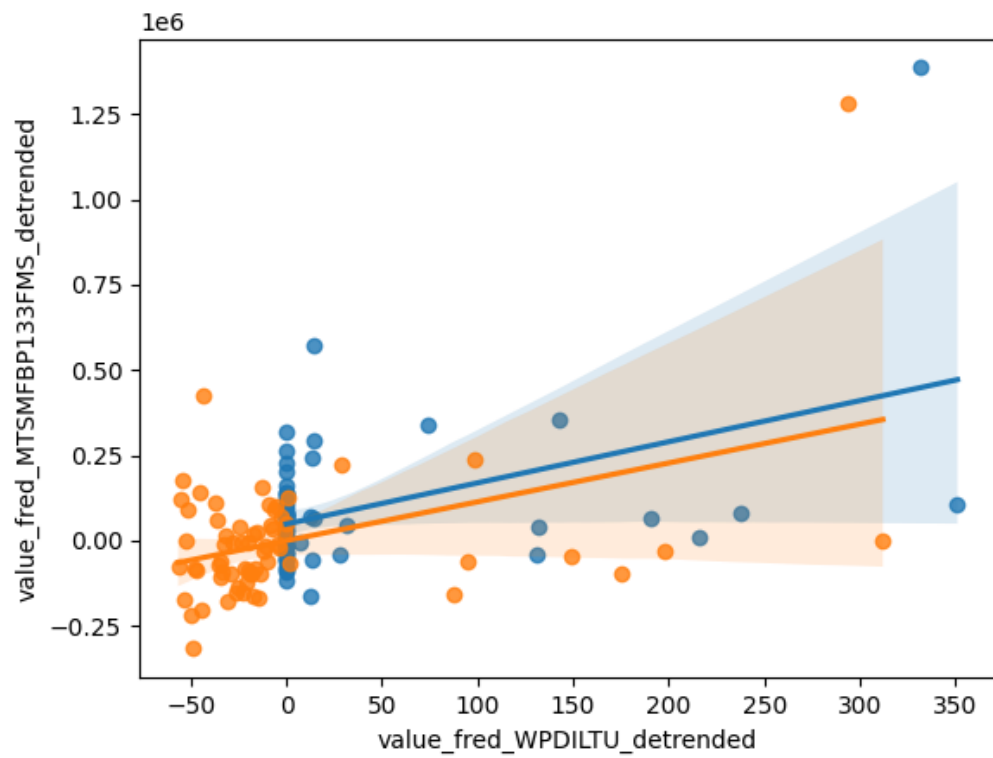


Figure 2: Detrended Regression Plot for 2026-09-07

3 Date: 2026-09-08

Series ID: EMISSCO2TOTVTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_COGANOPIMEPS	R-squared:	0.606
Model:	OLS	Adj. R-squared:	0.585
Method:	Least Squares	F-statistic:	29.22
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	3.24e-05
Time:	15:27:44	Log-Likelihood:	-111.42
No. Observations:	21	AIC:	226.8
Df Residuals:	19	BIC:	228.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-146.4574	51.777	-2.829	0.011	-254.827	-38.088
value_fred_EMISSCO2TOTVTTCOUSA	0.1533	0.028	5.406	0.000	0.094	0.213

Omnibus:	3.022	Durbin-Watson:	1.167
Prob(Omnibus):	0.221	Jarque-Bera (JB):	1.530
Skew:	0.625	Prob(JB):	0.465
Kurtosis:	3.429	Cond. No.	8.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

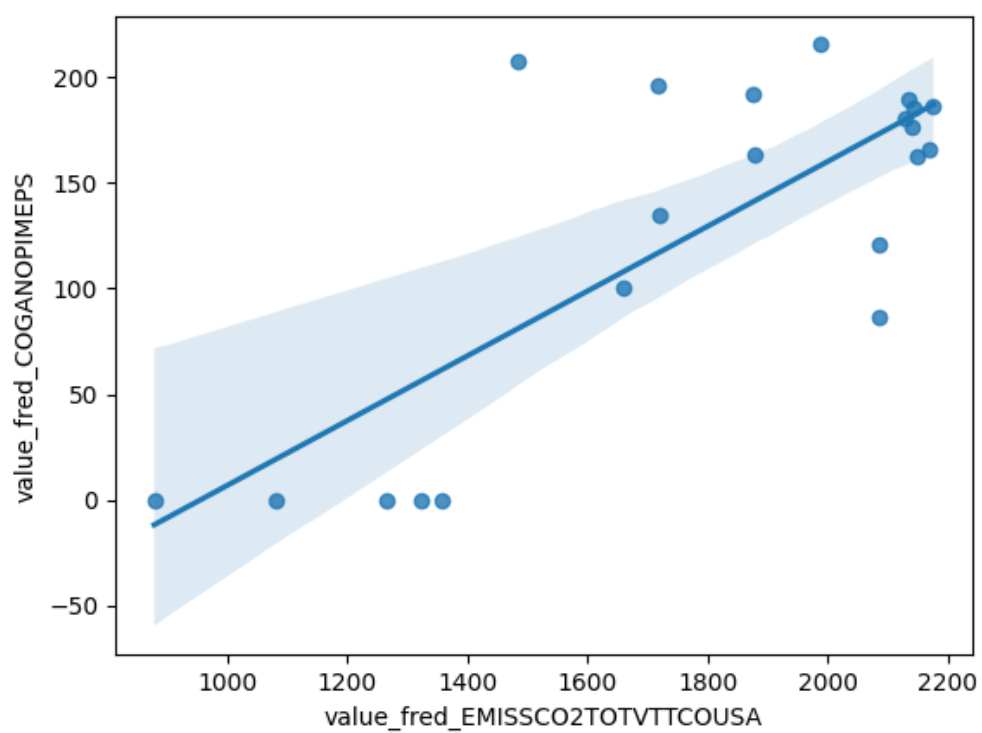


Figure 3: Raw Regression Plot for 2026-09-08

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_COGANOPIMEPS_detrended	R-squared:	0.559
Model:	OLS	Adj. R-squared:	0.536
Method:	Least Squares	F-statistic:	24.09
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	9.76e-05
Time:	15:27:44	Log-Likelihood:	-108.14
No. Observations:	21	AIC:	220.3
Df Residuals:	19	BIC:	222.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	-6.734e-12	9.567	-7.04e-13	1.000	-20.023	20.023
value_fred_EMISSCO2TOTVTTCOUSA_detrended	0.3032	0.062	4.908	0.000	0.174	0.432

Omnibus:	4.692	Durbin-Watson:	1.541
Prob(Omnibus):	0.096	Jarque-Bera (JB):	2.578
Skew:	0.726	Prob(JB):	0.276
Kurtosis:	3.915	Cond. No.	155.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

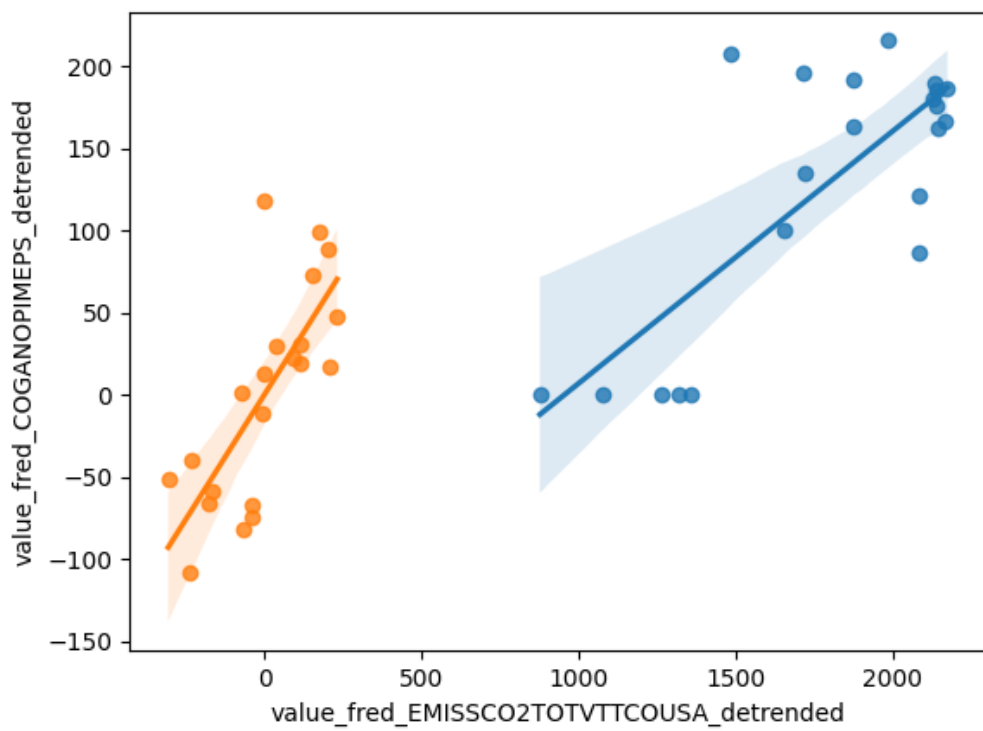


Figure 4: Detrended Regression Plot for 2026-09-08